VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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MINI PROJECT REPORT
(BEC586)

Submitted in the partial fulfillment of the requirement for the award of degree

Voice Control Home Automation With Smart Door Lock

BACHELOR OF ENGINEERING IN
ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by:

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UNDER THE GUIDANCE
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BENGALURU- 560091 2025-2026

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CERTIFICATE

This is to certify that the mini project work report entitled "Voice Control Home Automation With Smart Door Lock" is a bonafide work carried out by RAKSHITHA JAIN S (1EW23EC125), RUQUIYA KHANUM (1EW23EC130), SNEHA CHAUHAN (1EW23EC155), VARSHA BASAVARAJA GOTUR (1EW23EC172), in partial fulfillment for the award of degree of Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-2026. This mini project work report has been approved as it satisfies the academic requirement in respect of mini project work prescribed for Bachelor of Engineering Degree.

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DECLARATION

We **RAKSHITHA JAIN S (1EW23EC125), RUQUIYA KHANUM (1EW23EC130), SNEHA CHAUHAN (1EW23EC155), VARSHA BASAVARAJA GOTUR (1EW23EC172)** students of V semester Bachelor of Engineering in the Department of Electronics & Communication Engineering of East West Institute of Technology, Bengalore-560091, hereby declare that the mini project entitled **"Voice Control Home Automation With Smart Door Lock"** has been carried out by us under the supervision of Internal Guide **Mrs. Shilpa V , Assistant Professor**, Department of Electronics & Communication Engineering, EWIT, Bangalore submitted in the fulfillment of the courser requirement for the award of the degree of Bachelor of Engineering in Electronics & Communication Engineering of Visvesvaraya Technological University during the academic year 2025- 2026.

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ABSTRACT

IoT-enabled smart home automation system utilizing the ESP32 microcontroller to enhance home management through real-time remote control. The proposed system integrates various sensors, including temperature, humidity, motion (PIR), light, and raindrop sensors, to automate tasks based on environmental conditions, thereby improving energy efficiency and convenience. Additionally, the system incorporates an RFID-based door locking mechanism for enhanced security and supports voice commands for hands-free operation, catering to individuals with mobility limitations. A web interface allows users to monitor and control connected devices remotely via Wi-Fi, offering a seamless and userfriendly experience. Emphasis is placed on secure communication protocols to safeguard user data, while a modular architecture ensures scalability for future enhancements. This solution demonstrates the potential of IoT technologies to create. energy-efficient, secure, and accessible smart home systems, addressing modern demands for sustainable living.

Keywords- Voice Control, Home Automation, Smart Door Lock, IOT, Wireless Control, Security System, Voice Recognition, Access Control, Smart Home, Microcontroller(ESP32).

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CHAPTER:1

INTRODUCTION

The rapid advancements in Internet of Things (IoT) technology have revolutionized various aspects of modern living, with smart home automation being one of the most prominent applications. Smart home systems provide homeowners with the ability to remotely monitor and control household devices such as lighting, doors, and environmental sensors through smartphones, voice commands, or other connected devices. These systems enhance convenience, improve energy efficiency, and strengthen security, making them an integral part of sustainable and intelligent living solutions. A smart home leverages IoT to seamlessly integrate devices and sensors, creating an interconnected network that automates tasks based on user preferences and real-time environmental data. This approach not only reduces manual effort but also enables energy optimization by ensuring appliances operate only when required. For instance, sensors can monitor temperature, humidity, or motion to adjust lighting, heating, or cooling systems automatically, leading to significant energy savings and improved environmental impact. Furthermore, smart home automation addresses critical concerns in home security.

Features like RFID-based access control, motion detection, and realtime notifications allow users to monitor their homes remotely and respond to potential threats effectively. Such systems also cater to the needs of individuals with limited mobility by incorporating voice recognition for hands-free operation, enhancing accessibility and user convenience. The growing urbanization and rising energy demands underscore the importance of integrating energy-efficient solutions in residential settings. The adoption of IoT-enabled smart home systems is a step towards achieving this goal while also enhancing the quality of life for users.

1.1 PROBLEM STATEMENT:

In modern households, the lack of an integrated, affordable, and intelligent automation system creates challenges in ensuring security, convenience, and energy efficiency. Traditional home appliances require manual operation, which can be difficult for elderly individuals or people with mobility limitations. Similarly, conventional door lock systems relying on physical keys pose risks such as loss, duplication, and unauthorized access.

Existing automation solutions often lack real-time monitoring, environmental sensing, and seamless voice-based control.

Therefore, there is a need for a smart home system that combines IoT technology, sensorbased automation, secure RFID door access, and voice command functionality to provide users with a safe, accessible, and energy-efficient living environment. This project addresses these challenges by developing an ESP32-based home automation system capable of remote monitoring, intelligent control, and enhanced security.

1.2 OBJECTIVES:

The aim of this project is to design and develop an efficient IoT-based smart home automation system that can be controlled using voice commands, RFID authentication, and web-based monitoring. This system will automate household tasks, enhance home security, and offer hands-free operation, providing users with greater convenience, accessibility, and control over their living environment.

- **Smart Home Automation:** To automate the control of home appliances such as lights, fans, and doors using IoT-enabled ESP32 for seamless operation with minimal human intervention.
- **Voice-Controlled Operation:** To integrate a voice recognition interface that enables users to control home devices hands-free, improving accessibility and ease of use.
- **Secure Access Control:** To implement an RFID-based smart door locking system that allows only authorized users to enter, ensuring enhanced home security.
- **Environmental Monitoring:** To use sensors such as DHT11, PIR, LDR, and raindrop modules for monitoring temperature, humidity, motion, and rain to automate actions and improve energy efficiency.
- **Remote Monitoring and Control:** To develop a web-based dashboard that allows users to monitor and operate home devices remotely through Wi-Fi connectivity.
- **User-Friendly Interface:** To create an intuitive and responsive interface for both web and voice control, ensuring accessibility for all users, including elderly and physically challenged individuals.

- **Energy Efficiency:** To optimize power consumption by automating appliances based on sensor readings, thereby reducing unnecessary energy usage.
- **Scalability and Flexibility:** To design a modular system architecture that supports future upgrades, additional sensors, and extended home automation features.

1.3 MOTIVATION:

- **Time-Saving:** The increasing demand for automated home systems motivates the need for a solution that reduces manual effort in controlling household devices.
- **Convenience:** Voice-controlled and app-based smart automation offers a hands-free experience, catering to users who seek modern and convenient smart home solutions.
- **Security Enhancement:** Growing concerns about home safety encourage the development of smart locking mechanisms using RFID technology to prevent unauthorized access.
- **Technological Advancement:** The rapid evolution of IoT, wireless connectivity, and embedded systems provides an opportunity to design intelligent systems that improve everyday living.
- **Accessibility:** Voice-operated systems provide greater accessibility for individuals with mobility challenges, helping them control home appliances independently.
- **Energy Conservation:** Automating appliances based on environmental data promotes responsible energy usage and reduces wastage.
- **Smart Living and Comfort:** Modern lifestyles demand systems that enhance comfort by automatically adjusting lighting, security, and environmental conditions.
- **Sustainability:** Developing an IoT-based system prioritizes low power consumption and efficient resource usage, contributing to sustainable smart home development.

CHAPTER 2:

LITERATURE SURVEY

[1] H. Patel, D. Gupta, and R. Singh, "Domicile - An IoT Based Smart Home Automation System," in IEEE International Conference on Computational Intelligence and Networks (CINE), 2021.

It focuses on providing efficient remote monitoring and control of household appliances through wireless communication technologies. Their work highlights the integration of WiFi–enabled microcontrollers with sensors such as temperature, motion, and light detectors to automate essential home functions while enhancing user convenience and safety. The system supports real-time feedback and mobile-based control, enabling users to operate home devices from any location. The authors emphasize the benefits of IoT in improving energy efficiency by automating appliances based on environmental conditions and optimizing power usage. The study also demonstrates the scalability and cost-effectiveness of IoT architectures, making them suitable for modern smart homes. This research establishes a strong foundation for developing advanced smart home solutions by showcasing how IoT technologies can transform traditional households into intelligent, adaptive, and secure living environments.

[2] Jampana, J. G., Praneeth, A. J., Devi, J. H., & Rani, P. Smart Home Automation System with Status Feedback Based on ESP32 and IoT. Proceedings of the 7th International Conference on Innovations and Research in Technology and Engineering, (2022).

Jampana J. G., Praneeth A. J., Devi J. H., and Rani P. (2022) propose an ESP32-based smart home automation system that emphasizes real-time status feedback and reliable device control through IoT connectivity. Their work utilizes the ESP32 microcontroller for its builtin Wi-Fi capabilities, enabling seamless communication between sensors, appliances, and a cloud-connected user interface. The study highlights the importance of two-way communication, where users not only send commands to home devices but also receive instant feedback regarding their operational status, enhancing system reliability and user confidence. The authors demonstrate how the integration of sensors such as motion detectors, temperature sensors, and light sensors can support automated decision-making for improved energy efficiency and comfort. The research also focuses on designing a userfriendly dashboard that allows remote monitoring and control, making the system suitable for modern smart living applications. This study reinforces the potential of ESP32 and IoT technologies in creating intelligent, responsive, and efficient home automation systems.

[3] A. Koushal, R. Gupta, and F. Jan, "Home Automation System Using ESP32 and Firebase," in Proceedings of the IEEE International Conference on Smart Systems and IoT Innovations (ICSSI), 2023.

A. Koushal, R. Gupta, and F. Jan (2023) present an IoT-driven home automation system that integrates the ESP32 microcontroller with Google Firebase to enable seamless real-time device monitoring and control. Their work focuses on cloud-based interaction, where Firebase acts as a centralized database for storing sensor data and synchronizing appliance states across multiple user devices. The study highlights the advantages of combining ESP32's Wi-Fi capabilities with Firebase's fast data update features, ensuring low-latency communication and reliable automation performance. The authors demonstrate the system's ability to manage various household appliances through mobile applications, offering features such as remote access, real-time notifications, and secure data handling. Additionally, the architecture supports voice or app-based control, enhancing convenience and accessibility for users. This research underscores the growing significance of cloudintegrated IoT systems in developing scalable, user-friendly, and intelligent smart home environments.

[4] R Arunadevi, J Lurdhumary, R Ramya, M S Vinotheni, YuvikaShree S, Swathi D R, K Mariyamma- "Iot based smart home automation using esp32" 2025 International Conference on computing and communication technologies (ICCCT),2025.

R. Arunadevi, J. Lurdhumary, R. Ramya, M. S. Vinotheni, YuvikaShree S., Swathi D. R., and K. Mariyamma (2025) propose an IoT-based smart home automation framework utilizing the ESP32 microcontroller to achieve efficient monitoring and control of residential appliances. Their work highlights the ESP32's integrated Wi-Fi capabilities, low power consumption, and cost-effectiveness, making it an ideal platform for scalable IoT applications. The system incorporates sensors for environmental monitoring, including temperature, humidity, and motion detection, enabling automated responses based on realtime conditions. The authors emphasize the modular design of the system, which supports voice-based control, app-based interaction, and real-time data visualization for enhanced user convenience. The study demonstrates successful implementation of appliance control through cloud connectivity, ensuring remote accessibility and improved home security. This research contributes to the growing field of smart home technologies by showcasing how ESP32-based architectures can provide reliable, energy-efficient, and user-friendly automation solutions suitable for modern households.

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CHAPTER 3:

METHODOLOGY

The design of an IoT-based smart home automation system involves the integration of multiple hardware and software components to achieve efficient device control, real-time monitoring, and secure access management. The system is structured around the ESP32 microcontroller, which acts as the central hub, interfacing with various sensors and actuators to automate home functions. The hardware design consists of several essential modules. The ESP32 microcontroller is selected for its dual-core processing capability, integrated Wi-Fi and Bluetooth connectivity, and low power consumption, making it ideal for IoT applications. The DHT11 sensor is incorporated to monitor temperature and humidity, allowing automated climate adjustments based on real-time data. Motion detection is handled by a PIR sensor, which activates lights or security alerts upon detecting movement.

A photoresistor sensor enables ambient light-based automation, adjusting indoor lighting to optimize energy efficiency. Furthermore, a raindrop sensor is included to detect precipitation and trigger the automatic closing of windows or activating alerts for homeowners. For security and controlled access, an RFID module is implemented, allowing doors to be locked and unlocked based on authorized tag recognition. This enhances security while reducing reliance on traditional keys. Additionally, an LCD display is integrated to provide real-time system status updates, such as temperature, humidity, and access logs.

As shown in Fig3.1, block diagram of ESP32 DEV Board incorporates a voice recognition module, enabling hands-free control of various appliances, improving accessibility for users with mobility limitations. On the software side, the system employs an embedded programming approach using Arduino IDE with C++ for firmware development. The WiFi protocol facilitates remote access via a web-based user interface hosted on the ESP32. Communication between sensors and actuators is managed through I2C, UART, and HTTP protocols, ensuring seamless data transfer and device coordination. The 5 modular design architecture allows future scalability, enabling additional devices and features to be integrated without major modifications to the core system.

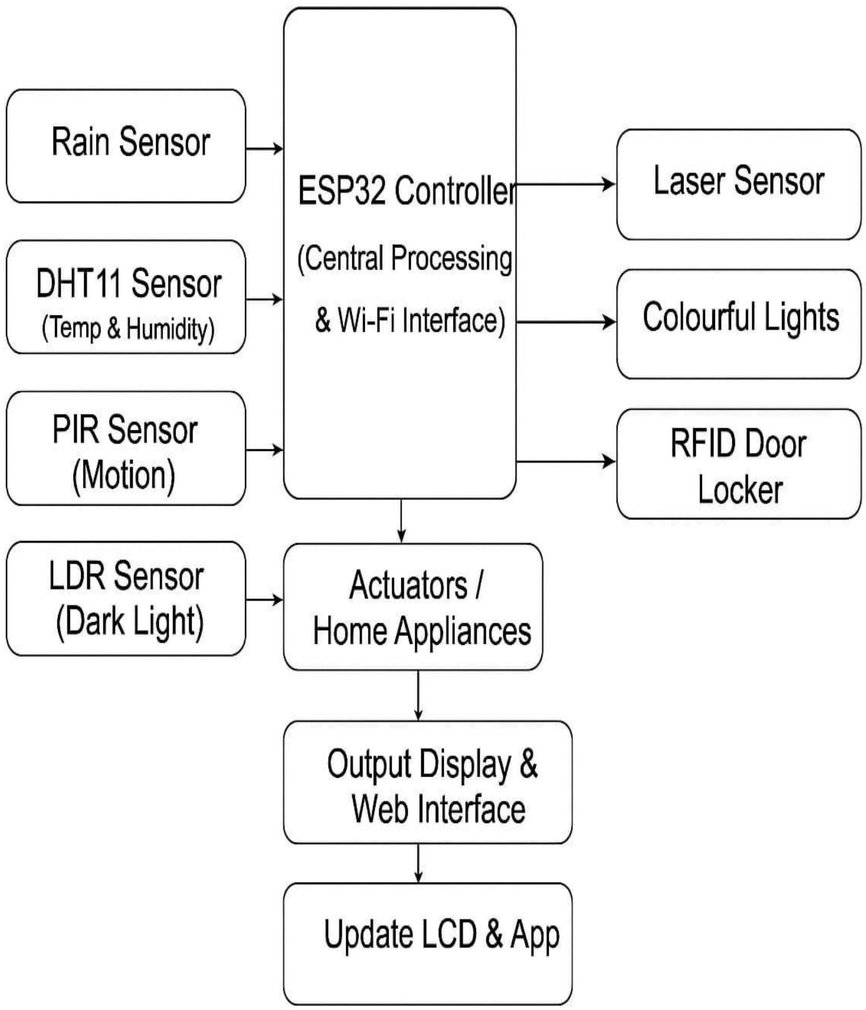

Fig.3.1 Block Diagram of ESP32 DEV BOARD

CHAPTER 4:

COMPONENTS REQUIRED

4.1 HARDWARE COMPONENTS REQUIRED

4.1.1 ESP32 MAX V1.0 CONTROLLER BOARD:

As shown in the Fig.4.1, the ESP32 MAX V1.0 is the main controller used in the smart home automation system. It has built-in Wi-Fi and Bluetooth, allowing remote control and voice operation. The board reads data from sensors like temperature, motion, light, and rain to automate different home functions. It also controls relays and the door lock system for switching appliances and securing the house.

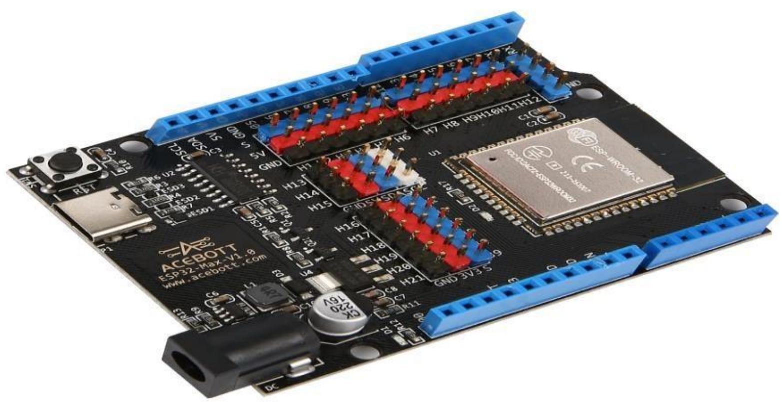

Fig.4.1 ESP32 MAX V1.0 CONTROLLER BOARD

The ESP32 supports RFID, LCD display, and voice modules to make the system safer and more user-friendly. It is programmed using the Arduino IDE and works as a small web server for real-time monitoring and device control.

4.1.2 BLUE LED MODULE:

As shown in the Fig.4.2, the Blue LED module is used as an indicator in the smart home system to show the status of different operations. When the ESP32 sends a signal, the LED lights up to indicate actions such as power ON, successful RFID access, motion detection, or system alerts. It helps users easily understand whether a device or function is active without checking the app or display.

Fig.4.2 BLUE LED MODULE

The module uses very low power and connects directly to the controller through simple digital pins. This makes it a reliable and clear visual indicator for various smart home notifications.

4.1.3 RAINDROP SENSOR:

As shown in the Fig.4.3, the raindrop sensor is used to detect rain in the smart home system. When water droplets fall on the sensor's surface, it sends a signal to the ESP32 to show that it is raining. This helps the system automatically perform actions like closing windows, turning off outdoor appliances, or giving alerts to the user.

Fig.4.3 RAINDROP SENSOR

The sensor is easy to connect and works with very little power. It provides quick and accurate rain detection, making the home automation system safer and more convenient during weather changes.

4.1.4 VOICE RECOGNITION MODULE V2.0:

As shown in the Fig.4.4, the Voice Recognition Module V2.0 allows the smart home system to understand and respond to spoken commands. When the user says a specific word or phrase, the module recognizes it and sends a signal to the ESP32 to perform actions like turning ON lights, opening the door, or controlling appliances.

Fig.4.4 VOICE RECOGNITION MODULE V2.0

It makes the system hands-free and convenient, especially for elderly or differently-abled users. The module is easy to connect and works using simple digital communication with the controller. Overall, it adds comfort and smart accessibility to the home automation project.

4.1.5 LASER MODULE:

As shown in the Fig.4.5, the laser module is used in the smart home system mainly for security and motion detection purposes. It produces a small, focused light beam that can be paired with a sensor to detect if someone passes through the beam.

Fig.4.5 LASER MODULE

When the beam is broken, the module signals the ESP32 to trigger an alert, turn on lights, or activate a security response. It is easy to connect, uses very low power, and provides a clear and reliable way to monitor movement in restricted areas. This makes the home automation system safer and more responsive.

4.1.6 DHT11 TEMPERATURE AND HUMIDITY SENSOR:

As shown in the Fig.4.6, the DHT11 sensor is used to measure the temperature and humidity inside the home. It sends this information to the ESP32 so the system can make smart decisions, like turning on a fan, adjusting ventilation, or showing the values on a display.

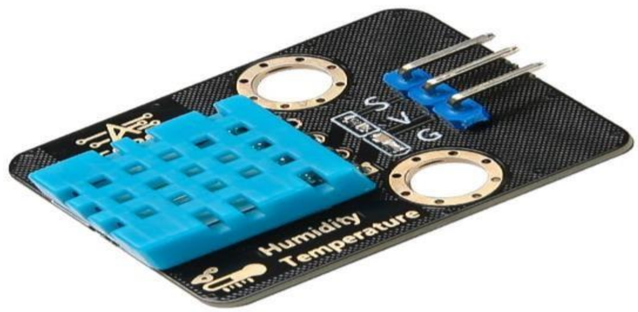

Fig.4.6 DHT11 TEMPERATURE AND HUMIDITY SENSOR

The sensor is easy to use, needs very little power, and gives accurate readings for basic home monitoring. By providing real-time weather conditions inside the house, the DHT11 helps the automation system maintain a comfortable and healthy environment.

4.1.7 PIR MOTION SENSOR:

As shown in the Fig.4.7, the PIR motion sensor is used to detect human movement in the smart home system. When a person walks near the sensor, it senses the body heat and sends a signal to the ESP32. This helps the system automatically turn on lights, trigger alarms, or send alerts for security.

Fig.4.7 PIR MOTION SENSOR

The sensor is easy to use, works with very low power, and gives quick and reliable motion detection. It helps make the home safer and also saves energy by activating devices only when someone is present.

4.1.8 RFID MODULE+ IC KEYCHAIN:

As shown in the Fig.4.8, the RFID module and IC keychain are used together to provide secure door access in the smart home system. When the user taps the IC keychain near the RFID reader, it identifies the unique card ID and sends it to the ESP32.

Fig.4.8 RFID MODULE+ IC KEYCHAIN

If the ID is authorized, the system unlocks the door; if not, it stays locked. This makes entry safe, quick, and keyless. The module is easy to connect and works reliably, helping improve home security without needing traditional keys.

4.1.9 BUTTON MODULE:

As shown in the Fig.4.9, the button module is used as a manual control switch in the smart home system. When the user presses the button, it sends a signal to the ESP32 to perform actions like turning a device ON/OFF or resetting a function.

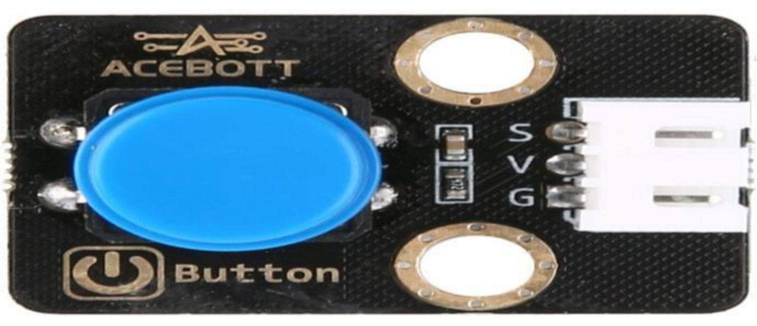

Fig.4.9 BUTTON MODULE

It acts as a backup control option when voice or app control is not preferred. The module is easy to connect, very reliable, and works using simple digital input. Overall, it provides a quick and convenient way to control the system manually.

4.1.10 PHOTORESISTOR SENSOR:

As shown in the Fig.4.10, the photoresistor sensor, also called an LDR, is used to measure the amount of light in the surroundings. When it becomes dark, the sensor's value changes and sends a signal to the ESP32, allowing the system to automatically turn ON lights or activate night-mode functions.

Fig.4.10 PHOTORESISTOR SENSOR

In bright light, it helps switch OFF unnecessary lighting to save energy. The sensor is easy to use, low-cost, and responds quickly to changes in light. This makes the smart home system more efficient and automatic based on day-night conditions.

4.1.11 LCD 1602 MODULE:

As shown in the Fig.4.11, the LCD 1602 module is used to display important information in the smart home system. It can show values like temperature, humidity, RFID access status, or system messages received from the ESP32.

Fig.4.11 LCD 1602 MODULE

This helps users easily see what is happening in real time without needing a phone or computer. The display is simple to connect using a few wires and works on low power.

Overall, it provides a clear and convenient way to monitor the system's data and activities.

4.1.12 6 AA BATTERY HOLDER:

As shown in the Fig.4.12, the 6 AA battery holder is used to supply portable power to the smart home system or its modules. It holds six AA batteries securely and provides a stable voltage output to run components when an external power supply is not available.

Fig.4.12 6 AA BATTERY HOLDER

This makes the system more flexible and useful even during power cuts. The holder is easy to connect with wires and ensures safe and organized battery usage. Overall, it helps keep the circuit powered reliably and conveniently.

4.1.13 SERVO SG90 9G:

As shown in the Fig.4.13, the Servo SG90 9G is a small motor used in the smart home system to move parts like the smart door lock. When the ESP32 sends a signal, the servo rotates to a specific angle, allowing the door to lock or unlock smoothly.

Fig.4.13 SERVO SG90 9G

It is lightweight, uses very little power, and is easy to control with just one signal wire. The servo provides precise movement, making it perfect for small mechanical actions in home automation. Overall, it adds reliable and accurate motion to the system

4.1.14 RBG LED STRIP:

As shown in the Fig.4.14, the RGB LED strip is a flexible light strip that produces different colors using red, green, and blue LEDs. It is controlled by the ESP32 to create colorful lighting effects in the smart home system.

Fig.4.14 RBG LED STRIP

The strip can change colors based on voice commands, sensor readings, or user settings. It helps in showing system status and also makes the project look attractive.

4.1.16 F-F 3P DUPONT WIRE:

As shown in the Fig.4.16, the F-F 3P Dupont wire is a small connecting cable used to join different electronic components without soldering. It has female connectors on both ends, making it easy to plug into sensor modules, the ESP32, or a breadboard.

Fig.4.16 F-F 3P DUPONT WIRE

This wire helps in creating neat and secure connections while building the circuit. Because it carries three wires together, it keeps the setup organized and reduces wiring mistakes. Overall, it is an essential component for making quick, clean, and reliable connections in our smart home automation project.

4.1.17 F-F 4P DUPONT WIRE:

As shown in the Fig.4.17, the F-F 4P Dupont wire is a connector wire used to link electronic parts without soldering. It has female ends on both sides, so it can easily plug into sensors, modules, and the ESP32. Because it has four wires joined together, it can carry more signals at once, keeping the wiring neat and clean.

Fig.4.17 F-F 4P DUPONT WIRE

This wire helps make the circuit stable and reduces the chance of loose connections. Overall, it is very useful for making quick, safe, and organized connections in our smart home automation project.

4.1.18 WOODEN BOARD:

As shown in the Fig.4.18, the wooden board is used as a strong base to hold all the project components in one place. It helps keep the ESP32, sensors, wires, and other modules fixed neatly so nothing moves or gets damaged.

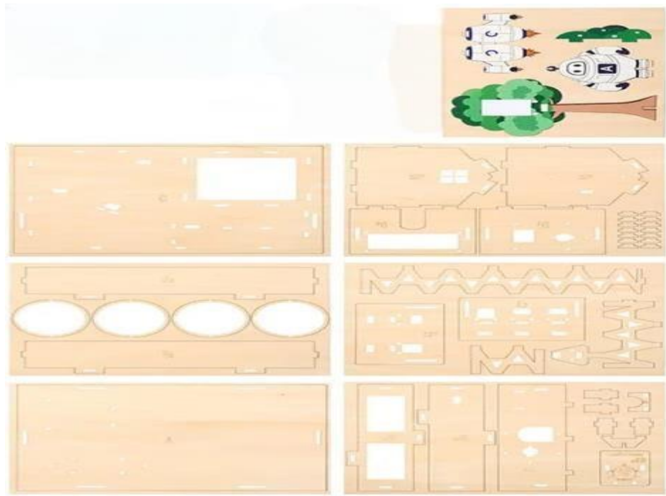

Fig.4.18 WOODEN BOARD

By mounting everything on the wooden board, the project looks clean, organized, and easy to display. It also makes the setup safer by preventing short circuits and wire tangling. Overall, the wooden board provides support and stability for the entire smart home automation system.

4.1.19 MIRROR KIT:

As shown in the Fig.4.19, the mirror kit is used in the project to give a clean and attractive look to the setup. It helps reflect light from LEDs or sensors, making the display clearer and more visible. The mirror also helps in neatly covering or arranging components so the project looks well-presented during demonstration.

Fig.4.19 MIRROR KIT

It does not affect the electronics but improves the overall appearance and organization of the system. Overall, the mirror kit makes the project look more professional and visually appealing.

4.1.20 ACRYLIC BOARD FOR SERVO:

As shown in the Fig.4.20, the acrylic board for the servo is used as a strong and clear base to hold the servo motor in place. It keeps the servo steady so it can move smoothly without shaking. Because acrylic is light, strong, and transparent, it makes the setup look neat and professional.

Fig.4.20 ACRYLIC BOARD FOR SERVO

Mounting the servo on the acrylic board also helps prevent damage and ensures accurate movement for door locking or any mechanical action. Overall, it provides support, safety, and a clean appearance to the servo motor setup in the project.

4.2 SOFTWARE COMPONENTS REQURIMENT

4.2.1 ARDUINO IDE:

Arduino IDE is the primary development environment used for writing and uploading code to the ESP32 microcontroller. It provides an easy-to-use interface, built-in libraries, and board management tools required for compiling C/C++ programs. With support for serial monitoring, debugging, and library integration, the Arduino IDE simplifies the development of IoT-based applications. It allows seamless communication with sensors, actuators, and communication modules, making it ideal for building smart home automation systems.

4.2.2 ESP32 BOARD LIBRARIES:

The ESP32 board libraries include the necessary drivers, functions, and APIs that allow the microcontroller to communicate with Wi-Fi, Bluetooth, sensors, and other embedded peripherals. These libraries enable GPIO control, Wi-Fi connectivity, interrupt handling, and communication protocol management. By installing the ESP32 board package, the Arduino

IDE can compile and upload sketches specifically tailored for the ESP32 hardware, ensuring reliable interaction with all connected modules.

4.2.3 HTML, CSS, AND JAVASCRIPT:

HTML, CSS, and JavaScript are used to create the web-based user interface that allows remote monitoring and control of home appliances. HTML structures the interface, CSS enhances the visual appearance, and JavaScript enables dynamic functionality for real-time updates. AJAX is used to continuously fetch sensor data and update the dashboard without refreshing the page. This combination ensures that users can interact with the system smoothly through any browser or smartphone.

4.2.4 HTTP, I2C, AND UART PROTOCOLS:

The HTTP protocol manages the exchange of data between the ESP32 and the web server, allowing control commands and sensor values to be transmitted in real time. I2C is used for communication with devices such as LCD displays and certain sensors due to its two-wire efficiency. UART facilitates serial communication with modules like the RFID reader and voice recognition system, ensuring fast and reliable data transmission.

4.2.5 ESP32 WEB SERVER:

The ESP32 Web Server allows the microcontroller to host a fully functional control panel accessible through any browser. It processes user commands, updates device states, and displays real-time sensor readings. By embedding the web server directly on the ESP32, the system eliminates the need for external servers or cloud platforms. This enhances reliability, reduces latency, and ensures that users can interact with the home automation system instantly and securely.

4.2.6 SPEECH-TO-TEXT LIBRARY / VOICE PROCESSING MODULE:

The speech-to-text library or voice processing module is responsible for converting spoken commands into machine-readable instructions. It enhances accessibility by allowing handsfree control of appliances such as lights, fans, and door locks. This feature is especially beneficial for elderly users, individuals with disabilities, or situations where manual control is inconvenient. The module processes voice input and sends the appropriate command signals to the ESP32 for execution.

4.2.7 USB-TO-UART DRIVER:

The USB-to-UART driver enables the computer to detect and communicate with the ESP32 development board. Without this driver, uploading code or accessing the serial monitor would not be possible. It establishes a stable interface between the PC and the microcontroller, ensuring smooth firmware uploads and real-time debugging. This is a necessary software component for programming and maintaining the ESP32 throughout the development process.

CHAPTER 5:

ADVANTAGES AND APPLICATIONS

5.1 ADVANTAGES:

Voice-Controlled Smart Home Automation with Smart Door Lock includes:

- **Hands-free operation:** Users can control lights, fans, and door locks using voice commands.
- **Increased home security:** RFID-based smart door lock prevents unauthorized access.
- **Energy saving:** Sensors like LDR, PIR, and DHT11 help reduce electricity waste by automating devices.
- **Remote control:** Users can monitor and control home appliances from anywhere through Wi-Fi.
- **User-friendly system:** The interface is simple, making it easy for anyone to operate.
- **Automation improves convenience:** Tasks like turning on lights or closing windows happen automatically.
- **Supports elderly and disabled people:** Voice control helps users who have mobility issues.

- **Real-time monitoring:** Temperature, humidity, and motion status are shown instantly.
- **Cost-effective design:** Uses affordable IoT components like ESP32 and sensors.
- **Scalable system:** New devices and features can be added easily in the future.

5.2 APPLICATIONS:

- **Smart Home Management:** Enables remote and voice-based control of lights, fans, and doors using Google Assistant, Alexa, or Blynk app. Offers in-app control mode for manual operation when voice mode is not preferred.
- **Automatic Window Control System:** Raindrop sensor detects rain and automatically closes windows, preventing water damage. Temperature and humidity sensors regulate window operation to maintain indoor comfort and air quality.
- **Motion Detection and Security:** PIR (human motion) sensors detect movement and trigger laser or colorful lights as alerts. Can be integrated with an alarm or camera system for enhanced home surveillance.
- **Dark Light Adaptive System:** Photosensitive (LDR) sensors detect darkness and automatically switch ON LED lights. Helps in energy saving by turning off lights when natural light is sufficient.
- **RFID-Based Smart Door Locking:** Allows secure RFID card-based entry for residents and authorized users. Prevents unauthorized access and provides multilayered security along with voice and app control.
- **Decorative and Safety Lighting System**: Laser and colorful lighting used for smart ambience control or security indication. Lights can respond dynamically to environmental conditions (like motion or dark mode).
- **Dual Control Modes (Voice and App):** Voice Mode: Enables hands-free control via virtual assistants. In-App Mode: Offers mobile app-based control for all components (lights, door, window). Both modes provide flexibility and accessibility for different users.
- **Personalized Home Automation**: System adapts to human presence using motion sensors. Lights, fans, or doors automatically adjust based on detected activity.

- **IoT-Based Environmental Monitoring:** Monitors temperature, humidity, and light intensity using ACEBOTT IoT sensors. Data can be viewed on a mobile dashboard, enabling real-time home environment analysis.

CHAPTER 6:

RESULT

As shown in Fig.6.1, the proposed Voice-Controlled Home Automation System with Smart Door Lock was successfully designed, implemented, and tested using the ESP32 microcontroller. The system accurately performed real-time monitoring and automatic control of various home appliances and security components. The integration of temperature, humidity, motion, LDR, and raindrop sensors allowed the system to respond intelligently to environmental changes, enabling automatic lighting, window control, and security alerts. Voice commands were successfully recognized by the voice-processing module, enabling hands-free operation of lights, fans, and the smart door lock. This feature significantly improved user convenience, especially for elderly or physically challenged individuals.

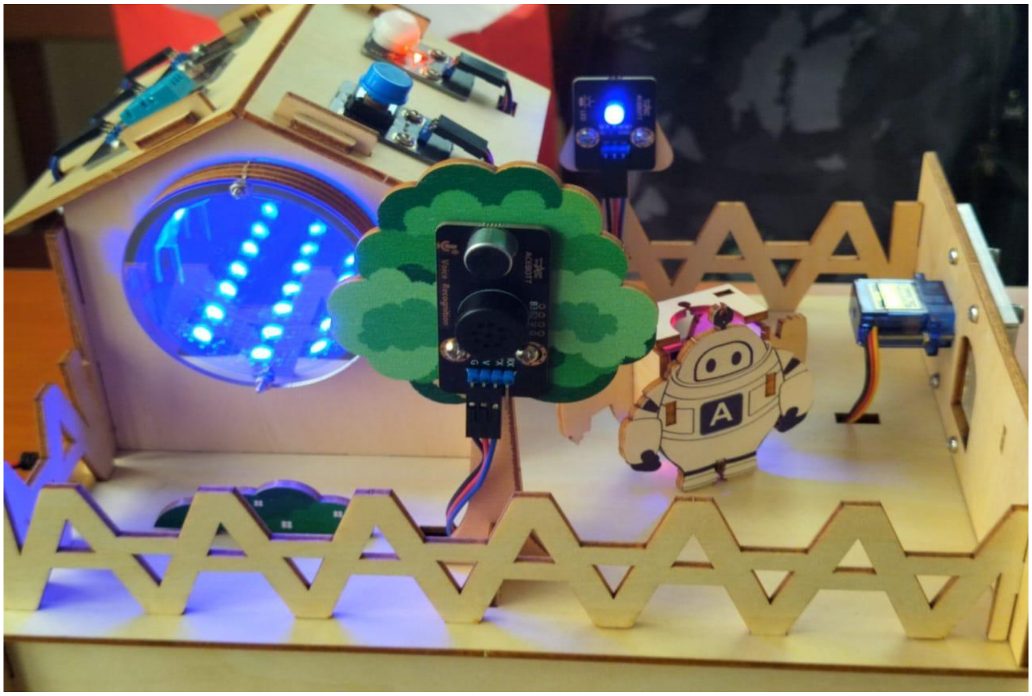

Fig.6.1 Voice Control Home Automation With Smart Door Lock

As shown in Fig.6.2, This image shows the smart door lock mechanism used in the home automation setup. The model includes a wooden or cardboard door structure integrated with an RFID-based locking system. A blue RFID or NFC tag is being used to trigger the unlocking process, which is handled by an electronic mechanism such as a servo motor fixed inside the door. This demonstrates secure access control, where only authorized tags can unlock the door, making it suitable for smart home entry systems.

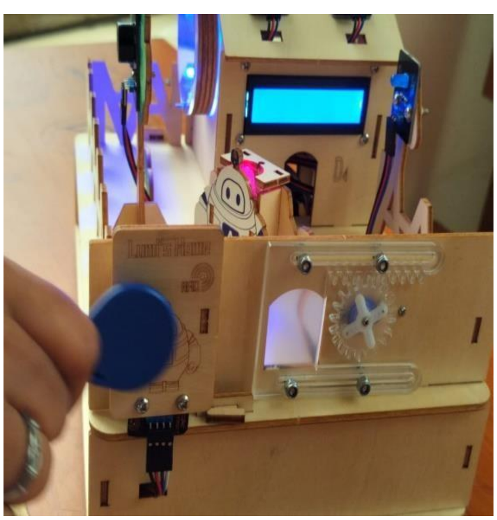

Fig.6.2 Door Lock System **Fig.6.3 LCD Screen (Temperature &Humidity)**

Fig.6.4 Close-up view of sensors

As shown in Fig.6.3, this picture displays an LCD screen mounted on the model to show real-time environmental data. The display indicates temperature and humidity values, which are likely obtained from sensors such as the DHT11 or DHT22. The screen also shows the system mode or status information. This setup highlights the data-monitoring feature of the smart home system, enabling users to keep track of indoor climate conditions.

Fig.6.4 provides a detailed look at multiple sensors installed around the smart home prototype. The cube-shaped structure holds various modules, including PIR motion sensors, ultrasonic distance sensors, gas and air quality sensors, and light-detection components. The illuminated blue LED at the center indicates active sensor operation. This arrangement demonstrates how different sensing units work together to monitor the environment, detect movement, enhance security, and automate home operations.

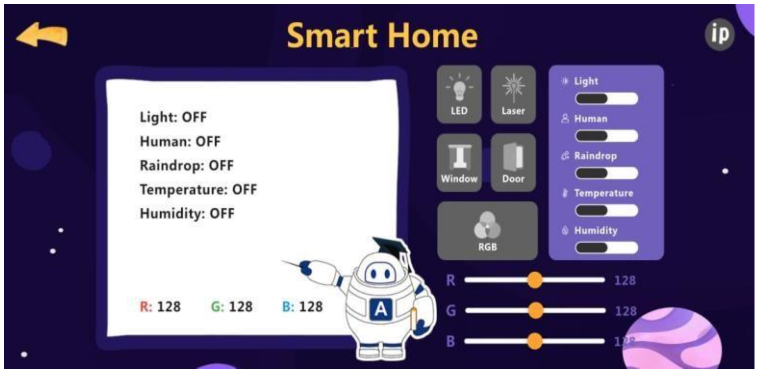

Fig.6.5 App control for Home Automation

Fig.6.5 shows, that the app mode allows the user to control the smart home system easily using a mobile phone. Through the app, lights, door, window, laser, and RGB lights can be turned ON or OFF using simple switches. The app also shows the status of sensors like human motion, rain, temperature, and humidity in real time. RGB sliders help change the color and brightness of lights smoothly. The system responds quickly to app commands through Wi-Fi. Overall, the app mode makes the smart home system simple, fast, and convenient to use.

CHAPTER 7:

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION:

The Voice-Controlled Smart Home Automation with Smart Door Lock project proves how modern IoT technology can greatly improve daily living. By integrating the ESP32 with sensors like PIR, LDR, DHT11, and an RFID module, the system provides automatic control, security, and real-time monitoring in a simple and efficient way. Voice commands make the system easy to use, especially for elderly people or users with mobility difficulties. The smart door lock ensures secure entry, while automation features help save electricity and reduce human effort. The web-based interface allows users to monitor and control their home from anywhere, making the system highly practical and reliable. Overall, this project demonstrates a powerful, scalable, and cost-effective solution for smart homes, showing how technology can enhance comfort, safety, and energy efficiency in a modern household.

7.2 FUTURESCOPE:

The Scope of future work in development of voice control home automation with smart door lock includes:

- The system can be expanded to support more smart appliances and rooms in the house.
- AI-based voice recognition can be added for faster and more accurate voice control.
- Security can be improved by adding face recognition or fingerprint authentication.
- Cloud storage can be used to save sensor data and analyze it for home energy optimization.
- A dedicated mobile app can be developed for better control, notifications, and user experience.
- Integration with smart assistants like Alexa and Google Home can make the system even more flexible.
- Automation rules can be enhanced so devices automatically adjust based on user habits and daily routines.

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